

A Demonstration Paper for Research-Integrity Auditing

Abstract. This synthetic paper contains deliberately planted statistical and image errors so that an automated integrity auditor can be evaluated end to end. It is not a real study.

Methods and Results. Participants were randomly assigned to a treatment or control group. The treatment improved performance, and the difference was statistically significant, $t(19) = 2.1$, $p < .01$.

Participants also rated their satisfaction on a 7-point Likert item with integer responses from 1 to 7. The treatment group ($n = 28$) reported a mean satisfaction of 5.19 ($SD = 1.30$), while the control group ($n = 28$) reported a mean of 4.00 ($SD = 1.10$).

Of the 200 enrolled participants, 38% (62 of 200) achieved remission. The cohort comprised 120 women and 75 men.

Figure 1 and Figure 2 show representative microscopy fields; Figure 3 shows an additional field from a separate sample.

Meta-analysis. We pooled four studies in a random-effects meta-analysis of treatment versus control. The pooled odds ratio was 3.20 (95% CI 2.40 to 4.27). Included studies: Almeida 2018 OR 2.10 (1.40-3.15); Becker 2019 OR 1.50 (1.05-2.14); Cho 2020 OR 2.40 (1.50-3.84); Diaz 2021 OR 1.20 (0.80-1.80).

Figure 1 (left) and Figure 2 (right)

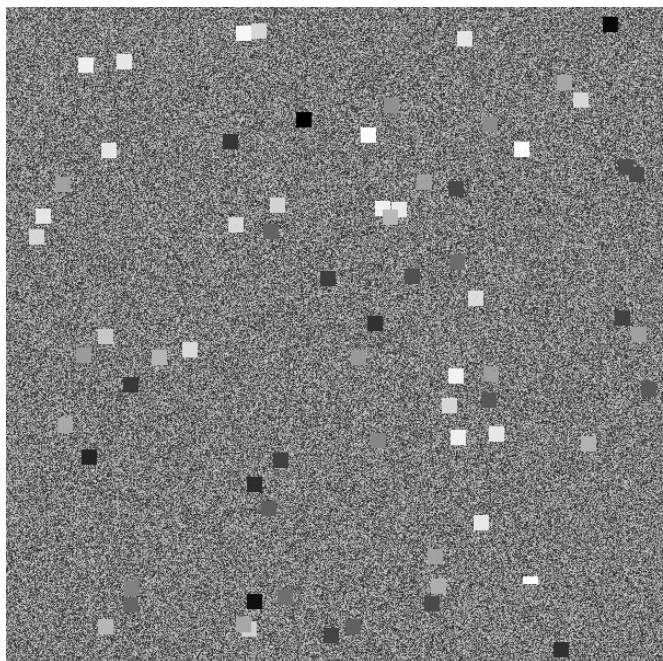
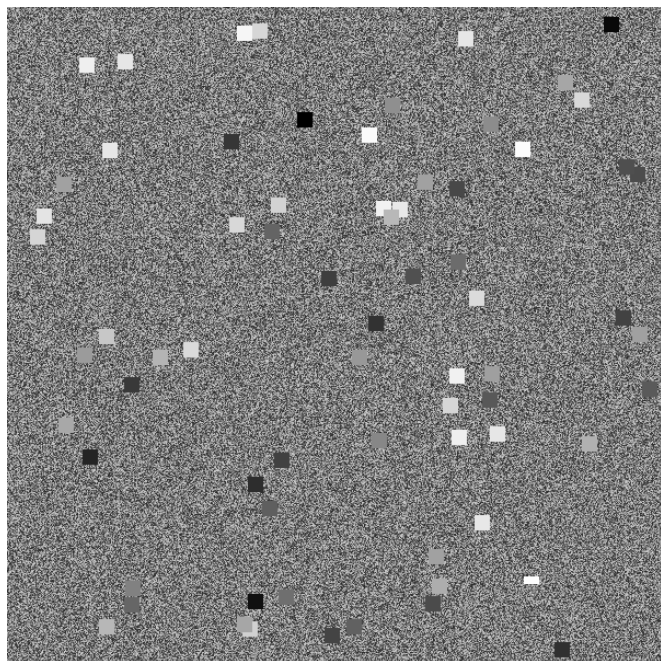


Figure 3

